

Passive Radar: Past, Present, and Future Challenges

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EVOLUTION OF PASSIVE RADAR TECHNOLOGY

The history of passive bistatic radar dates back to the Daventry Experiment, conducted by Sir Robert Watson-Watt in February 1935. To demonstrate the feasibility of this technology, he employed the shortwave signals emitted by the BBC Empire transmitter at Daventry to detect a Heyford bomber aircraft up to a few kilometers [1], [2], [3], [4], [5]. The first operational passive radars (PR) were the German “Klein Heidelberg” receivers, operating in 1943 to detect aircraft based on the emissions of the British “Chain Home” radars, installed along the south and east coasts of England [6], [7]. After World War II, several cooperative bistatic radars were developed; however, the attention of the radar community was mostly focused on active monostatic radar technology.

A revival of attention toward PR occurred between 1980s and 1990s, with pioneering contributions that demonstrated the feasibility of this technology and contributed to this renewed interest, such as the one from Griffiths and Long [8], followed by the study of Howland [9], both using the analogue TV transmissions from the Crystal Palace for aerial target detection. This is also reflected in the trend reported in Figure 1, which shows the number of research publications appeared in the technical literature per year on PR topics. This analysis has been conducted on the Scopus website, including papers that contain the words “passive radar,” “passive bistatic radar,” “passive covert radar,” (PCR) or “passive coherent location” in the paper title or among the keywords.

The renewed interest was mostly driven by the need to react to the increasingly congested electromagnetic spectrum, caused by the proliferation of communication services [10]. Over the past 40 years, the technical literature

has attested many attempts to exploit several transmitters as sources of opportunity to detect and localize targets, and PR technology gradually became a very active research area [11], [12], [13], [15], [16], [17], [18], [19]. During its 50 years of activity, the IEEE Aerospace and Electronic Systems Society witnessed this growth with a leading role, being most of the contributions published on its journals and presented at its major conferences.

First efforts and interests in the PR field focused on the possibility to enable detection capabilities without the need of a dedicated transmitter [8], [11], [12], [13]; as research in this field progressed, more radar functionalities were considered to provide PR systems with location, tracking, imaging, and classification capabilities in different scenarios (air, sea etc.) and for different applications (both military and civil). This process was accompanied by an intense study and development of the processing techniques required to ensure the abovementioned functionalities: from the more conventional processing techniques for cross-correlation and cancellation of direct signal and clutter to techniques that explore multidimensional data (e.g., spatial, frequency, polarimetric diversity...) and that are able to manage very long integration times.

The absence of a dedicated transmitter and the possibility to build a radar receiver by using commercial off-the-shelf components allowed many universities and research institutions to develop experimental systems; therefore, research activity at the theoretical level has been nicely complemented by validation carried out by means of ad hoc experimental campaigns.

From an industrial point of view, developments needed to face the fact that PR is a type of sensor different from classical active radar systems. The characteristic of PR systems of relying on transmitting facilities managed by third parties and on waveforms normally not optimized for radar purposes, probably motivated a delay in the involvement of the industry. Niche applications, however, could take advantage of the properties typical for PR, like the low cost, the possibility of covert operation, the lack of additional electromagnetic pollution, the low elevation coverage, and potentially long integration times. This motivated industrial developments and today operational systems are on the market.

The first commercial PR prototype using FM-radio broadcast emissions was developed by Lockheed Martin [20] and

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referred to as “Silent Sentry.” This was followed by further products from industries and research institutes. Without claiming to be exhaustive, among them are the long-range PR system TwinVis from Hensoldt (Germany); the AULOS PCR from Leonardo (Italy); the Ground Alerter 100 radar from Thales (France); the Passive Surveillance ESM Tracker (PET) VERA-NG from ERA (Czech Republic); the PARASOL, LORA, and SABBIA systems from Fraunhofer Institute (Germany), in some cases sharing the use with FFI (Norway); and the System PaRaDe from Warsaw University of Technology (Poland). The interested

reader is also referred to [2], [4], [19] for more complete descriptions.

Academic activity and industrial developments considered the use of different types of illuminators, both ground-based transmitters [analogue FM-radio, digital audio broadcast (DAB), digital video broadcast-terrestrial (DVB-T), cell-phone base stations, and WiFi], and satellite-borne transmitters (GNSS, DVB-S). As shown in Figure 2, depending on the considered operative scenario, some illuminators of opportunity (IOs) were demonstrated to be more suitable than others.

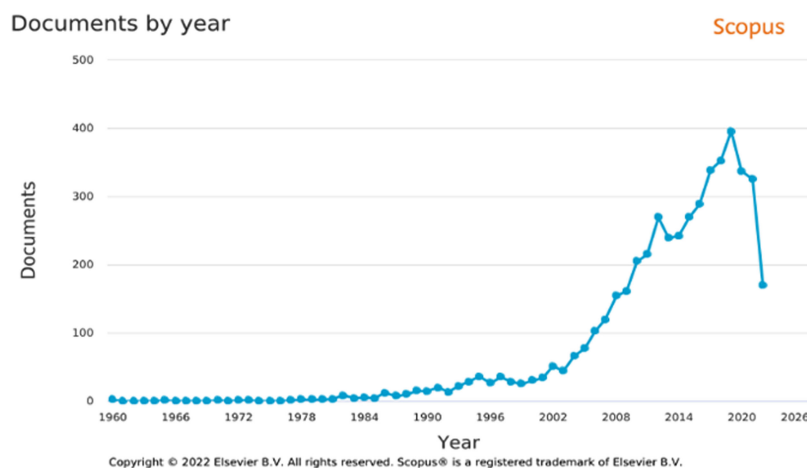


Figure 1.
Number of research documents per year on PR.

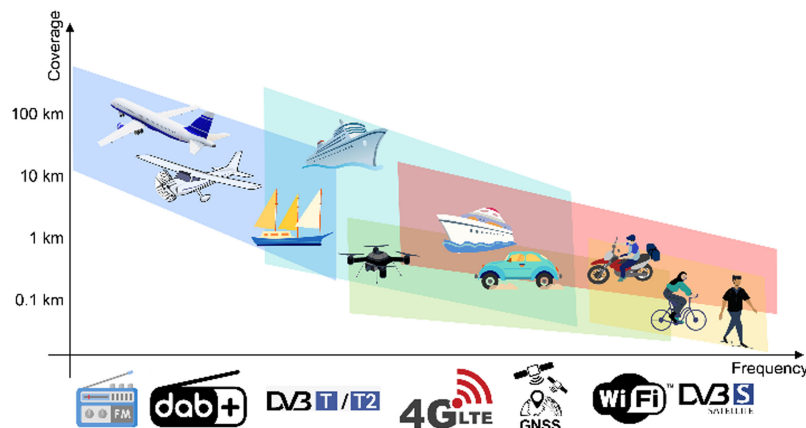


Figure 2.
Illuminators of opportunity for PR.

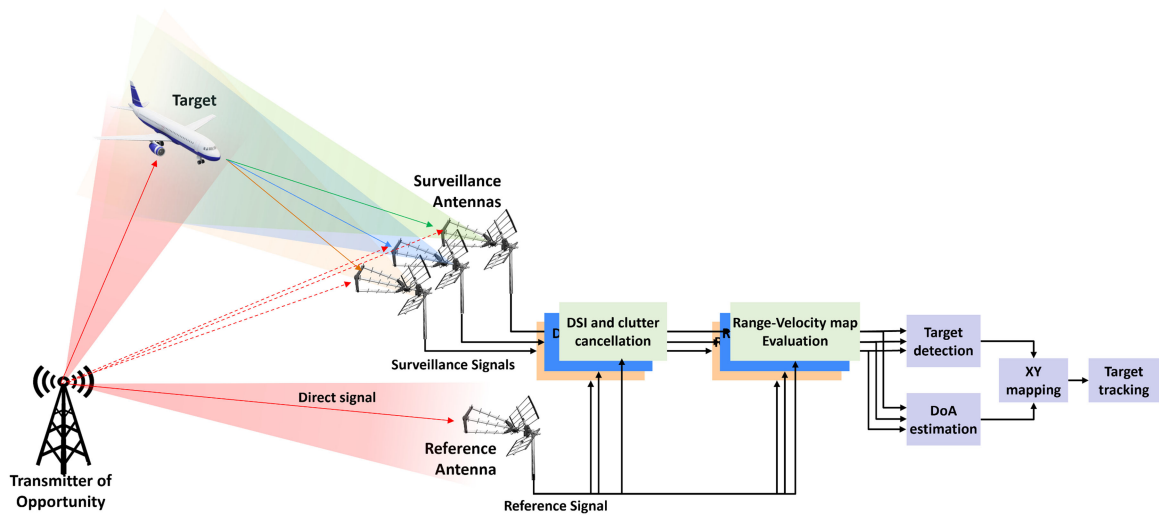


Figure 3.

Operative scenario for ground-based transmitters and main PR processing blocks.

Broadcast services transmitters, including ground-based both analog and digital audio/video broadcasting services, are suitable for long-range surveillance of aerial and maritime targets. Moving toward higher frequency and wider bandwidth signals (such as those of satellite transmitters or base stations for networking services), PR systems can be employed for shorter range monitoring applications to detect small drones, vehicles, and people or for remote areas surveillance.

In this article, PR is considered from the perspective given by the class of exploited transmitters of opportunity: for each class (ground-based transmitters, satellite-based transmitters, and wireless network services), strengths and weaknesses are discussed and results for different fields of application illustrated.

GROUND-BASED TRANSMITTERS OF OPPORTUNITY FOR BROADCAST SERVICES

Terrestrial transmitters of opportunity for broadcast services have been the most popular and widely used in PR technology so far. In fact, the extensive coverage and great continuity offered by broadcast transmitters, e.g., FM radio, DAB, or DVB-T, make these IOs especially suitable for long-range surveillance applications, which require high reliability and robustness to the conditions of the operating environment. For these reasons, PR technology relying on such IOs has been the subject of many research activities and lead to the development of an effective signal processing scheme.

A typical operative scenario is sketched in Figure 3. Here, the passive system employs a reference antenna steered toward the transmitter acquiring the direct signal (which acts as the reference signal for matched filtering) and multiple surveillance antennas pointed in the direction to be surveyed

receiving the reflected signal (due to the reflection of the transmitted signal from the target). Along with the sketch of the considered scenario, Figure 3 shows a block diagram of the typical signal processing scheme foreseen in a PR system. To facilitate the understanding of the following sections even for a nonexpert reader, in the following, we briefly describe the main processing stages; however, we refer the interested reader to the referenced technical literature for detailed explanations and additional insights.

- *Direct signal interference (DSI) and clutter cancellation:* This stage is aimed at removing from each surveillance signal the fraction of the direct signal emitted by the exploited IO (dashed red lines in Figure 3), as well as additional interferences and multipath contributions coming from the stationary scene. An effective DSI removal is crucial to avoid masking effects on the low-power target echoes. A common way to perform this stage is to subtract from the signal collected from each surveillance antenna the undesired interference signal, estimated by summing up delayed and weighted replicas of the direct signal, [21], [22], [23], typically acquired by the reference antenna.
- *Range-velocity map evaluation:* Once the interfering contributions have been removed, the key step in the PR processing chain is based on the evaluation of the cross-ambiguity function between the surveillance signal and the reference signal. This stage approximates a bank of matched filters (MF) tuned to different Doppler values of interest and a potential target echo appears as a peak at a given bin of the resulting bistatic range-velocity map. The number of employed Doppler filters is proportional to the selected coherent processing interval (CPI), which is usually quite large in PR, to extract the low-power target echoes from the competing disturbance as

well as to guarantee a high Doppler frequency resolution. For these reasons, several endeavors were presented in the technical literature, providing efficient implementations, e.g., the correlation FFT or the direct FFT algorithms; suboptimum implementations, e.g., the batches algorithm as well as alternative approaches to the conventional MF [21], [24].

- *Target detection:* As in active radars, the target detection process aims at deciding whether a target is present in a cell under test of the bistatic range–velocity map, based on a comparison with a threshold. Depending on i) the statistical distribution of the competing disturbance, ii) the availability of multiple surveillance channels, iii) the employed approach, both the detection test and the strategy to set the threshold change. A commonly employed solution is to use a threshold that adapts itself to the local disturbance characteristics to guarantee a constant false alarm rate. Once a detection is declared, an estimate of the target bistatic range and velocity will be available. However, depending on the employed waveform of opportunity, large targets might occupy more than one bin in the map, resulting in multiple detections for a single target; therefore, clustering algorithms might be used to reduce the number of overall detections and to streamline the subsequent stages.
- *X-Y mapping and target tracking:* When multiple surveillance channels are available, which separately underwent the previous signal processing stages, a direction-of-arrival (DoA) estimate can be extracted for each detected target. This additional measurement enables the localization of the target peak in the Cartesian plane. Finally, target tracking is applied [25], [26], by exploiting the target detections collected at multiple consecutive CPIs to identify potential tracks and further reduce the resulting number of false alarms. Depending on the employed configuration, target tracking can be performed in the bistatic range–Doppler domain, in the Cartesian plane or in both.

The effectiveness of the described processing scheme has been extensively tested, especially in aerial and maritime surveillance applications. In recent years, significant efforts have been dedicated to the improvement of the core processing blocks in terms of both computational complexity and achievable performance, and to increase the coverage, reliability, and continuity of operation. In addition, attempts have been made to expand the range of use of these systems as well as to reduce their cost in order to facilitate their use in a sensor network.

These activities required the modification and/or extension of the PR processing scheme (e.g., to cope with multidimensional data), and some recent results along this line are presented in the following sections. For instance,

the “Multidimensional Approaches for Aerial and Maritime Surveillance” section shows the advantage of properly exploiting the information diversity collected in multiple domains, such as frequency and polarization while in the “DVB-T-Based PR for Counter Drones Operations” section, we demonstrate the potentialities of DVB-T-based PR for the detection of small UAVs.

MULTIDIMENSIONAL APPROACHES FOR AERIAL AND MARITIME SURVEILLANCE

Among the advanced strategies recently developed to increase the target detection capability and the coverage of broadcast-based PR systems, an interesting opportunity is offered by multidimensional approaches that jointly combine the information diversity obtained in different information domains. For instance, polarimetric diversity can be easily obtained by employing dual-polarimetric antennas to collect the horizontal (H) and vertical (V) polarized versions of both the surveillance and reference signals [27], [28], [29]. Multipolarimetric approaches were shown to be useful in increasing the target detection capability and the ability to reject interfering contributions. Frequency diversity, on the other hand, is intrinsically offered by broadcast IOs that simultaneously transmit signals at multiple carrier frequencies [30], [31], [32] and it has been demonstrated to be useful to increase the robustness of the PR system to the time-varying characteristics of the employed waveform and of the propagation channel. Finally, these information diversity sources can be combined in a joint multifrequency and multipolarimetric adaptive detection scheme, as the ones proposed in [33], [34], and [35].

We show here the effectiveness of the approach proposed in [35] against real-world data, collected with an FM radio-based PR system for air traffic control (ATC). We report in Figure 4, the target detection results obtained for 30 consecutive CPIs of approximately 1 s each, with $P_{fa} = 10^{-4}$. Black lines represent the air-truth collected with an ADS-B receiver; gray dots are the raw PR detections while red dots represent the PR detections correctly associated with the air-truth. Specifically, the results obtained with the best performing single-frequency and single-polarimetric channel among those available are reported in Figure 4(a). Figure 4(b) shows the detection capability obtained with an approach that effectively leverages the information diversity obtained collecting, with dual-polarized antennas (H and V), signals simultaneously transmitted by the same IO at four different FM radio channels [35].

When using the conventional single-channel processing [see Figure 4(a)], the system guarantees good detection performance only up to approximately 50 km and obtains only few isolated correct detections at higher ranges.

The multidimensional approach in Figure 4(b), instead, yields a substantially higher number of detections in almost every track, up to 150 km, including those belonging to aircraft not equipped with transponders, as the sequence of

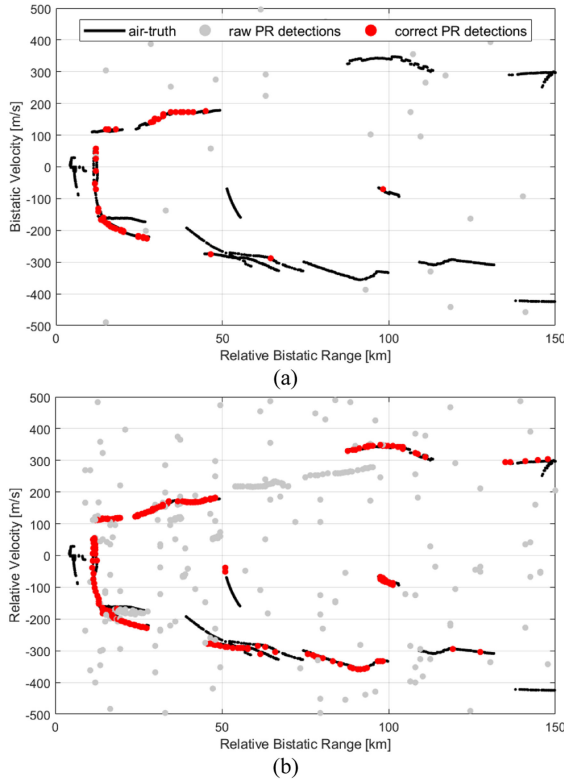


Figure 4.

Target detection results with (a) best single-frequency and single-pol channel; (b) multifrequency and multipolarimetric solution. Copyright © IEEE. All rights reserved. Reprinted with permission from [34].

consecutive plots between 50 and 100 km and at approximately 200–300 m/s. The obtained benefits are due to both a proper disturbance rejection at the single-frequency channel, obtained thanks to a fruitful exploitation of the polarimetric and temporal correlation, and to a noncoherent integration of the target echoes at the different carrier frequencies. We refer the reader to [35] for a detailed explanation of strategy and to gain further insights on the results.

DVB-T-BASED PR FOR COUNTER DRONES OPERATIONS

Along with the extensively demonstrated potentialities of DVB-T-based PR for both ATC and maritime surveillance applications, a challenging task that has been addressed in the recent years is the detection of small drones. In fact, differently from aircraft and vessels, drones are characterized by i) low velocities and unpredictable motion, ii) very small radar cross section. Therefore, efforts have been put into adapting the DVB-T-based PR processing chain, usually employed for long-range surveillance, to serve the application at hand. These adjustments have included the tailoring of the core processing blocks and their relevant parameters, e.g., the CPI, the update rates and time windows used for cancellation.

As an example, we consider in Figure 5 the data collected during an acquisition campaign at the military airport of Pratica di Mare, Rome, Italy, using the DVB-T-based AULOS PR sensor from Leonardo S.p.A [36], [37], employing two small drones (DJI Phantom 4 Pro and DJI Mavic Pro) as cooperative targets [see the acquisition geometry in Figure 5(a)]. The PR receiver was installed close to the airport runway, equipped with three surveillance antennas, displaced in a non-uniform linear array (NULA) configuration to provide good angular localization performance. Figure 5(b) and (c) reports the drone detection results obtained with $P_{fa} = 10^{-7}$ in the bistatic range-velocity plane and in the X - Y plane centered in the receiver location, respectively. The two different small UAVs, denoted by different colored lines and markers, can be detected and correctly localized up to a few kilometers, as extensively demonstrated in [37]. Note that a PR system based on DVB-T transmissions could be integrated in a multiband passive counter drone system, exploiting the complementary aspects of different IOs to accurately detect and locate small UAVs [38]. For instance, while DVB-T-based PR sensors guarantee the medium-range coverage of a few kilometers, DVB-S and WiFi-based PR sensors might offer higher localization accuracy in the shorter range. Finally, as will be illustrated in the “Wireless Network Service-Based Transmitters of Opportunity” section, the WiFi band operation potentially enables us to combine the PR operation with device-based localization approaches.

SATELLITE-BASED TRANSMITTERS OF OPPORTUNITY

While, from the beginning of PR, numerous studies and products focused on the exploitation of ground-based illuminators, only relatively recently attention has been paid to satellite illuminators. Noticeably, satellite-based PR present some relevant advantages with respect to the terrestrial-based ones: wider accessibility on the global scale and not reliance on potentially vulnerable infrastructures. Suitable candidates are communication satellites, such as Inmarsat and Iridium, broadcasting satellites in geostationary orbits, such as Astra and Eutelsat (DVB-S) and global navigation satellite systems (GNSS); each choice has its own merits and drawbacks.

The fundamental limitation related to the exploitation of these opportunity sources consists in the typically limited power budget. On the one hand, this limits the operative range to which these systems can arrive; on the other hand, this determines the need to develop adequate techniques to increase performance using very long integration times and/or appropriately capitalizing on the multidimensionality of the data (e.g., the spatial diversity offered by constellations, as in the case of GNSS).

Despite the abovementioned limitations, specific applications could greatly benefit from the key characteristics of satellite-based IOs: specifically, the continuous global (or at least

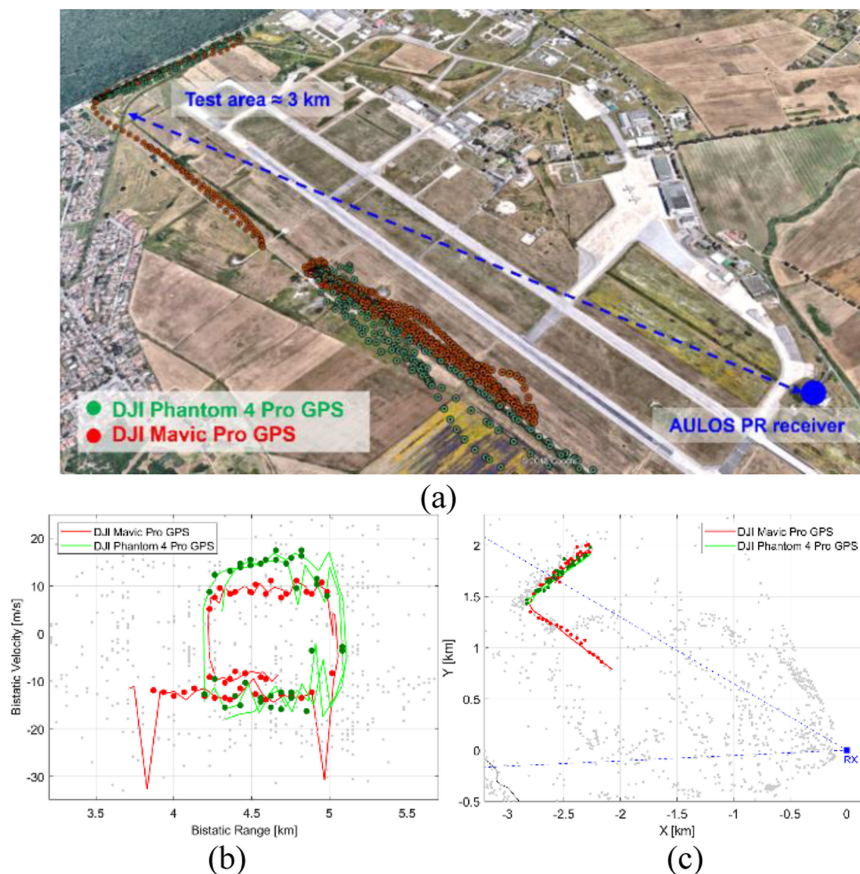


Figure 5.

DVB-T-based PR for drone detection: (a) acquisition geometry; target detection results in (b) bistatic range-velocity plane and (c) X - Y plane. Copyright © IET. All rights reserved. Reprinted with permission from [37].

continental) coverage could enable surveillance in remote areas not covered by ground-based transmitters and offers potentialities for short-range persistent area monitoring. The permanent availability of signal sources in remote areas is of great interest for maritime surveillance applications. Indeed, open sea areas cannot be reached by the radio waves emitted by ground-based transmitters but are in the scope of few satellite sources. This also applies to coastal or river navigation in scarcely populated remote areas. The system concept is sketched in Figure 6 and is based on the exploitation of satellite transmitters (GNSS in the figure) and passive receivers on the coast or on a moored buoy.

In this frame, the “GNSS-Based Ship Targets Detection” section describes maritime moving target detection by means of GNSS-based PR while the “GNSS-Based Ship Target Imaging” section considers the possible exploitation of the space-based sources (both GNSS and DVB-S) to image the detected target in order to enable classification/recognition for a complete surveillance. Finally, potentialities for short-range persistent area monitoring are shown in the “DVB-S-Based PR for Critical Infrastructure Protection” section where DVB-S is exploited for the detection of small-size

intruding vehicles approaching critical infrastructures. Despite the focus of this section is on the detection and imaging of moving targets, it is worth to highlight that satellite-based IOs have long been exploited for passive SAR imaging, particularly in the case of GNSS: the interested reader is referred to [39] and references therein for more on this topic.

GNSS-BASED SHIP TARGETS DETECTION

Navigation satellites are particularly suited for maritime/river surveillance: the generic constellation guarantees a persistent coverage of any point on the Earth (even at the poles) from six to eight satellites simultaneously; this number could rise up to 32 satellites with all the four global systems (GPS, GLO-NASS, Galileo, and BeiDou) in full capacity.

As example, Figure 7 shows a pictorial view of an experimental campaign carried out near Bonn, Germany, to monitor the traffic over the river Rhine [40], [41]. During the specific acquisition, four targets were in the field of view of the radar antenna: two passenger ships Godesia and Filia Rheni, moving away from the receiver, and the

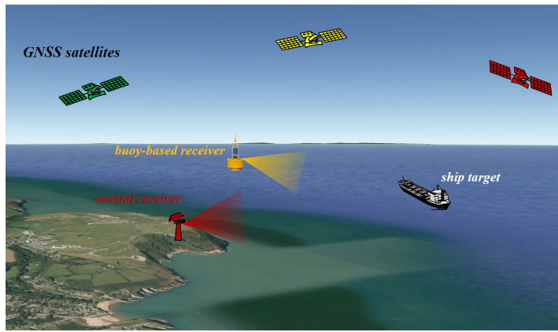


Figure 6.

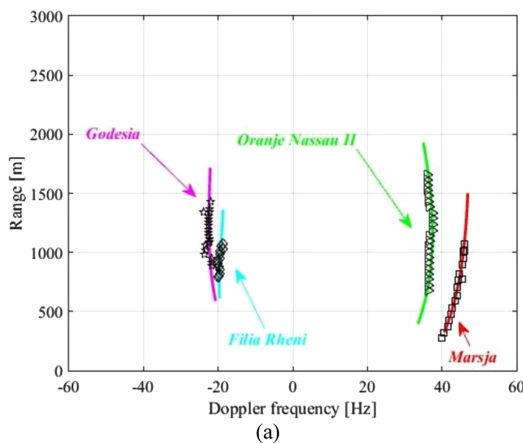
Operative scenario for satellite-based PR. Copyright © IEEE. All rights reserved. Reprinted with permission from [44].

two motor tankers Oranje Nassau II and Marsja, approaching; AIS tracks also shown in Figure 7. Two Galileo satellites were simultaneously tracked and separately exploited to monitor river traffic.

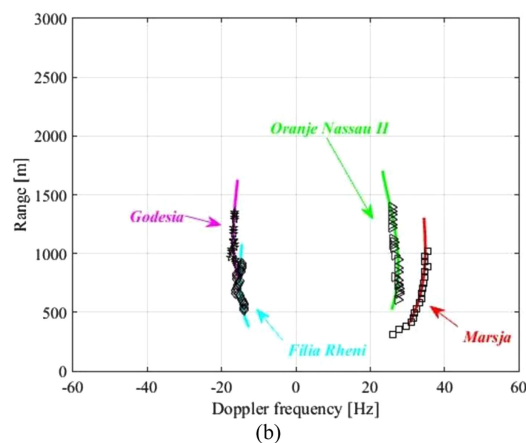
Figure 8 shows the results for both the resulting bistatic planes as obtained by jointly applying long integration time [42], [43], and track-before-detect techniques [41]. By comparing the detected tracklets to the AIS ground truth, a good agreement is observed for all the involved targets; in this case, the specific scenario and acquisition geometry limited the detection range since, for distances higher than 1500–1800 m, the target left the field of view of the receiving antenna. More in general, see for example [43], it has been demonstrated that a single bistatic link could enable detection up to few kilometers; considerable performance enhancement could arise by fully exploiting spatial diversity provided by the multiple satellites and constellations jointly allowing detection and localization [40].

GNSS-BASED SHIP TARGET IMAGING

Once ship targets have been detected, it could be of interest to image them to enable classification/recognition. To



(a)



(b)

Figure 8.

Rhine River detection results. (a) Sat 1. (b) Sat 2. Reprinted with permission from [41].

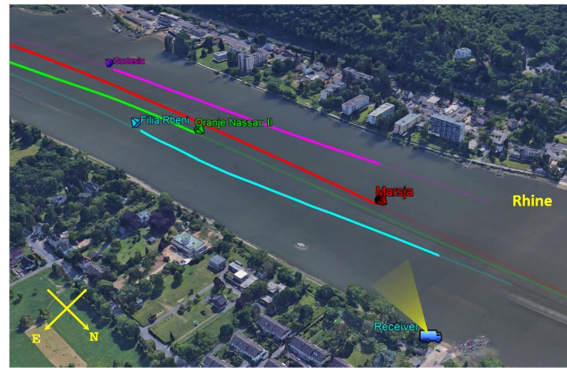


Figure 7.

Experimental campaign acquisition geometry. Reprinted with permission from [41].

this aim, ship target motion could be exploited to achieve cross-range resolution by means of inverse synthetic aperture radar (ISAR) techniques. Depending on the exploited bandwidth, the achievable ISAR products could be cross-range profiles or actual images.

As examples, Figure 9 shows results achievable exploiting GNSS [see Figure 9(a) and (b)] and DVB-S [see Figure 9(c)]. Specifically, Figure 9(a) and (b) shows the radar images of a commercial ferry (length 184 m and width 25 m) moving in the Plymouth Harbour, U.K., obtained deploying the prototype developed at the University of Birmingham and by exploiting the E5a-Q signal from a Galileo satellite [44]. Particularly, the five images in Figure 9(b) refer to five successive time windows, each of 15 s, being each window start time shifted by 3 s with respect to the previous one. Each image has been obtained by properly processing the corresponding time window, with coherent integration inside intervals 3 s each and noncoherent integration of the five CPIs. For comparison, Figure 9(a) shows the images related to a single CPI; the optical image is also reported.

DVB-S-based ISAR product is instead shown in Figure 9(c) and compared to the optical one: in this case, data were

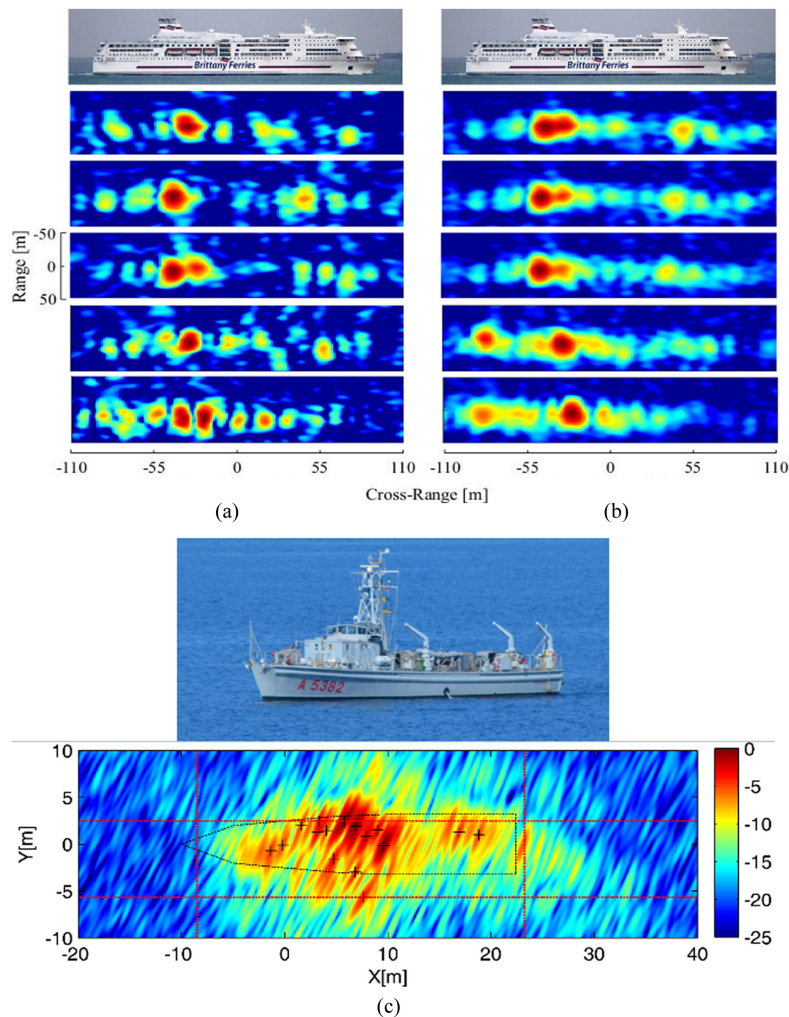


Figure 9.

(a) and (b) GNSS passive ISAR results; Copyright © IEEE. All rights reserved. Reprinted with permission from [44]. (c) DVB-S passive ISAR results. Copyright © IET. All rights reserved. Reprinted with permission from [45].

acquired in Livorno, Italy, deploying the experimental system SABBIA (developed at Fraunhofer FHR, Germany), which exploits broadcast satellite DVB-S(2) signals in Ku-band emitted from geostationary satellites, such as Astra [45].

The radar image of the involved cooperative target (length about 32.4 m and width about 6.4 m) is reported in the bottom subfigure with superimposed the extracted dominant scatterers (indicated by the black “+” marks in the figure) and the limits of the deck. For both cases, it is interesting to note how a correspondence between ship structure and radar images can be observed: this suggests that the fruitful extraction of extra information from the images (such as the length, width, etc.) can be possible, potentially enhancing the classification capability of the system. Further information on the imaged target could be obtained by properly exploiting multidimensional data, e.g., multiangle [46], [47] and multipolarization data [47], [48].

DVB-S-BASED PR FOR CRITICAL INFRASTRUCTURE PROTECTION

Another interesting application for satellite-based PR systems is the critical infrastructure protection, against small drones, cars, or people [49], [50]. Along with the common advantages of all the satellite-based transmitters, DVB-S waveforms are particularly suited for the application in hand since i) a single DVB-S channel typically has a frequency bandwidth of approx. 30 MHz, allowing a range resolution of about 10 m; ii) by operating in the Ku band, they also enable a high-velocity resolution; iii) based on the typical direct signal to noise and target signal to noise ratio (SNR) measured at the surveillance antennas, the DSI removal stage does not appear strictly necessary [50]; therefore, the processing scheme can be easily streamlined, enabling a

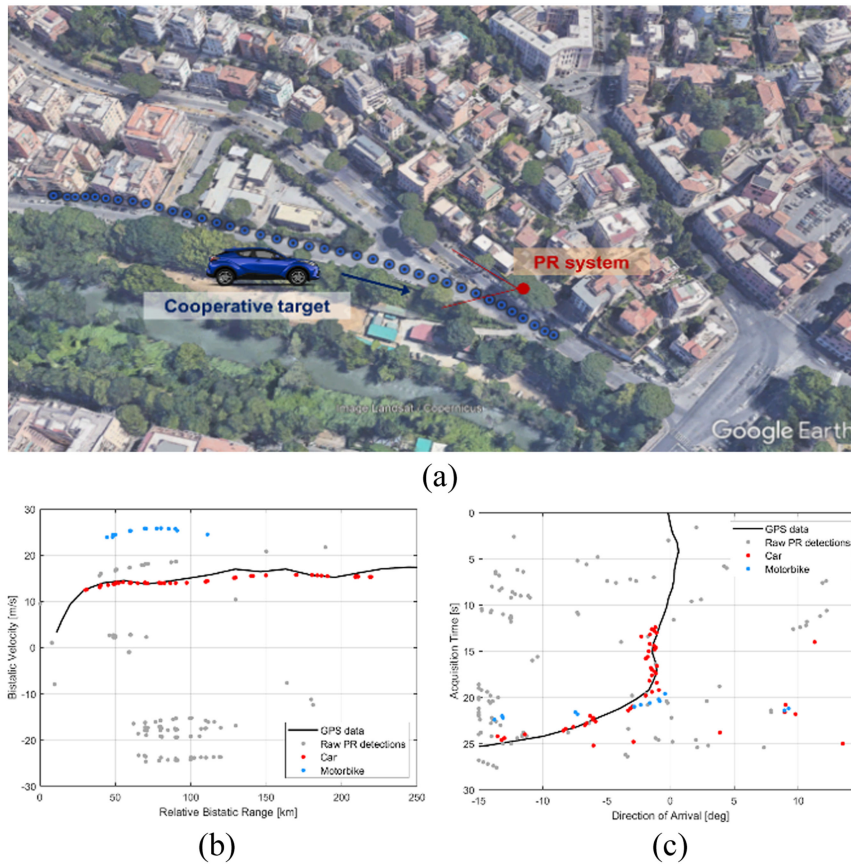


Figure 10.

(a) Acquisition geometry for DVB-S-based PR; (b) and (c) target detection results. Copyright © IEEE. All rights reserved. Reprinted with permission from [50].

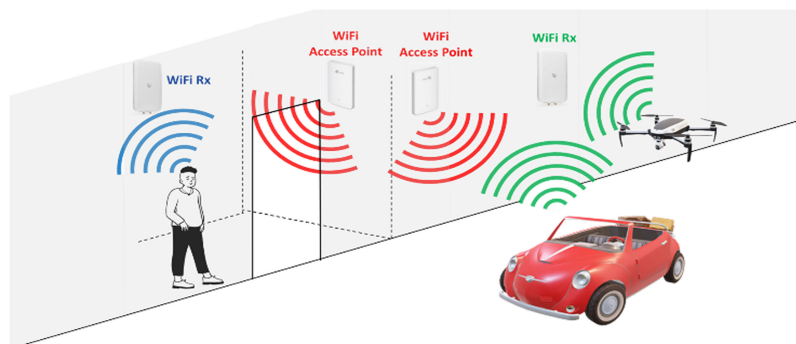


Figure 11.

Operative scenarios for Wi-Fi-based PR.

real-time operation, needed to provide prompt and frequently updated results.

The results of an illustrative experimental validation are reported in Figure 10. A test was performed with a DVB-S PR system equipped with a three-element array with nonuniformly spaced receiving elements, collecting the surveillance signals to enable angular localization and one reference antenna steered toward the EUTELSAT

33E satellite, emitting signals at 10.758 GHz. During the acquisition, a medium size car was employed as cooperative target and equipped with a GPS.

The GPS track is reported in Figure 10(a) and, as black line, in Figure 10(b) and (c), in the bistatic range–velocity and time–DoA plane, respectively. Figure 10(b) shows that the PR can detect the cooperative car when it reaches approximately 220 m of bistatic range. Moreover, during

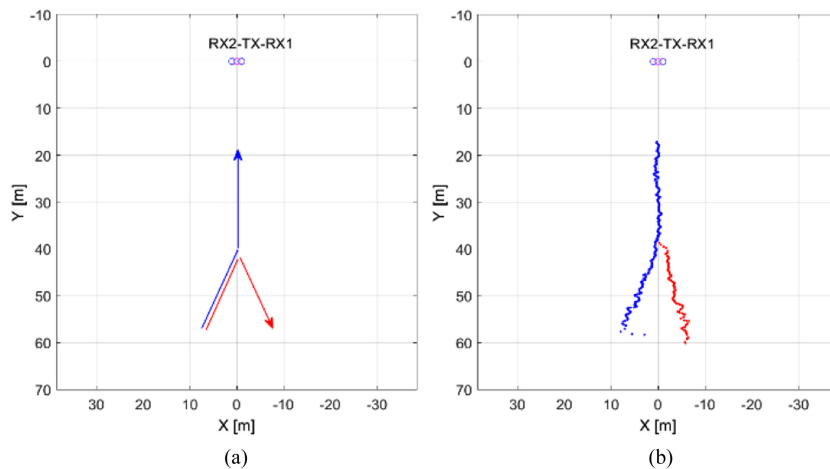


Figure 12.

Detection of human targets. (a) True trajectories. (b) PR detections in the XY plane. Copyright © IEEE. All rights reserved. Reprinted with permission from [56].

the entire acquisition, several horizontal sequences of continuous plots are visible in the range–velocity plane, which are likely to correspond to vehicles moving along the same or adjacent streets, either heading in the same direction as the cooperating car or in the opposite one. Among them, we denote as light blue dots a sequence of detections that correspond to a medium size motorbike, observed by the cooperating car driver shortly behind him. The same detections are then correctly localized in the time–DoA plane with good accuracy and only few ambiguous estimations, thanks to a proper NULA layout selected for the three surveillance antennas. Further insights on these results are reported in [50].

WIRELESS NETWORK SERVICE-BASED TRANSMITTERS OF OPPORTUNITY

In recent years, there has been a growing interest in developing monitoring solutions for indoor environments, either residential or commercial, due to the security and public safety issues as well as service matters. Real-world applications are often aimed at implementing smart environments and might include, for example, people location and navigation along buildings, detection of intruders, activity monitoring and e-healthcare, vehicle navigation or asset tracking. Along with video surveillance systems, various wireless technologies have been used for indoor monitoring among which infrared, IEEE 802.11 wireless LAN (WiFi), radio frequency identification, and ultrasonic.

In addition, PR technology and particularly WiFi-based PR has become an attractive solution due to several reasons.

- 1) WiFi access points are proliferating in modern buildings; therefore, WiFi-based PR sensors

represent a low-cost solution that leverages existing infrastructures and commercial hardware.

- 2) WiFi transmitters provide a limited coverage but potentially wide bandwidth and well-controlled signals useful for a very local parasitic exploitation (see Figure 11 for a sketch of the monitoring scenarios).
- 3) No extra signal has to be transmitted and this limits the energy consumption, prevents possible interference with pre-existing systems, and makes the sensor free from any issue related to human health; moreover, it is inherently nonintrusive and does not cause any discomfort or privacy issue.
- 4) WiFi-based PR does not require any cooperation from the target; therefore, compared to the above mentioned technologies, it is better suited for surveillance applications, such as intruder detection.
- 5) However, since some of the target of interest might carry devices that emit signals in the WiFi band, the implementation of passive WiFi sensors potentially enables the joint use of PR and device-based passive location approaches.

Despite the exploitation of the PR principle in short-range applications could appear as a simpler task, it brings in a number of challenges that must be addressed in order to benefit from its potential advantages.

In addition, the requirements on such sensors could be much more demanding to enable advanced capabilities, as required by progressively smarter environments.

Among the drawbacks of PR systems for indoor applications, the severity of the interference contributions (i.e., direct signal and multipath) is a key point. Therefore, effective signal processing techniques should be devised to enable continuous detection and accurate localization

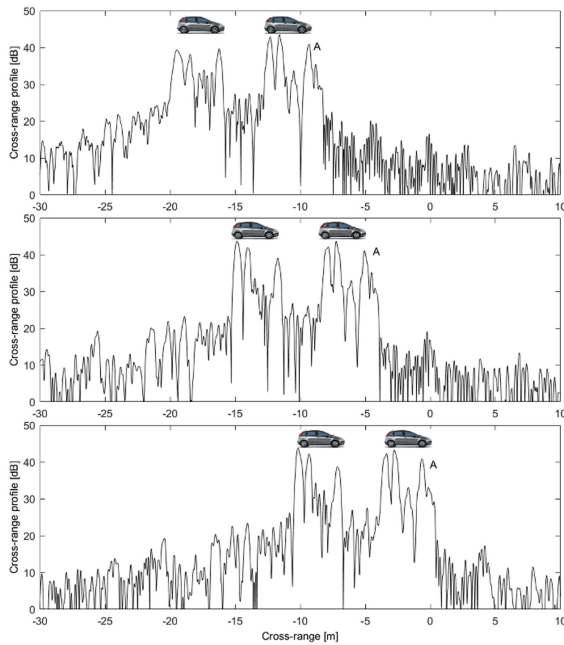


Figure 13.

Sequence of frames from cross-range profiling test employing two identical cars. Copyright © IEEE. All rights reserved. Reprinted with permission from [55].

of the targets of interest while keeping low the complexity of the resulting sensors and thus their cost.

In addition, the surveyed scene might be characterized by a high concentration of targets. Despite the most recent WiFi standards support very wide frequency bandwidth, the resulting radar range resolution might still be insufficient to discriminate closely spaced targets moving in the observed area or to distinguish different parts of their body. Therefore, alternative strategies should be considered to improve the resolution of these sensors.

In the following sections, practical examples of the use of WiFi-based PR are shown. Specifically, the potentialities and challenges mentioned previously are addressed and possible solutions are investigated and tested.

HUMAN AND VEHICULAR MONITORING

The effectiveness of WiFi-based PR for indoor monitoring has been demonstrated in several contributions appeared in the technical literature [51], [52], [53], [54], [55], [56], [57], [58], [59], [60], [61], [62]. Here, we report an example of the results obtained against human targets moving in a large room (see Figure 12). In the considered test, two people walk along partially overlapping paths [see actual trajectories in Figure 12 (a)]. In particular, they initially walk in the same direction with a constant separation of about 1 m; then, they abruptly change their heading, going toward opposite sides of the room. The WiFi-based PR exploits a commercial AP as IO (Tx), located at the origin of the Cartesian reference system. Moreover, two

surveillance antennas (Rx1–Rx2) displaced along the X -axis are employed to provide angular localization capability; they are installed in the proximity of the Tx in a quasi-monostatic configuration and are pointed toward the Y -axis to collect the echoes of targets moving in the room. The results of the WiFi-based PR are reported in Figure 12(b) using different colors for the different target tracks identified. It shows that the system can continuously detect and localize the two targets, with only small deviations with respect to their real trajectories. Note that the two targets can be distinguished only in the second part of the test because they move toward opposite directions, i.e., they show opposite Doppler frequencies.

The range information does not provide the required resolution capability as is clearly apparent from the first part of the test where the two targets are correctly detected but not distinguished. In fact, we recall that typical WiFi transmissions are characterized by a bandwidth between 20 and 40 MHz, which corresponds to a range resolution in the order of several meters. Improved resolution can be synthesized by adjoining multiple frequency channels transmitted by the same emitter or exploiting more recent IEEE 802.11 Standards, which support transmissions with a bandwidth of up to 160 MHz; however, the resulting improvement might still not allow discriminating targets in scenarios of limited size.

For this reason, target search and localization are often performed in alternative domains, such as angle or Doppler. In particular, for targets with a motion component in the cross-range direction, it is possible to achieve a higher (cross-range) resolution by applying appropriate ISAR processing schemes. Such approach provides the PR with the advanced capability to obtain high-resolution cross-range profiles of moving objects that could enable the automatic classification of designated targets. To this purpose, an effective ISAR scheme for cross-range profiling has been introduced in [55], tailored for the typical short-range surveillance scenarios of a WiFi-based PR.

Figure 13 shows an example of the results obtained for an experimental test performed against vehicular targets in a parking area, using a similar setup as the one employed in Figure 12, with the Tx and Rx mounted in a quasi-monostatic configuration. In the reported test, two identical cars (i.e., Fiat Punto Evo) were employed moving at about 4.5 m/s along a direction orthogonal to the Tx (Rx)-target line of sight, i.e., in a cross-range direction. Along their path, the two cars kept a separation of about 3–4 m between each other and they pass at 50 m from the Tx (Rx) at the point of closest approach.

Specifically, Figure 13 shows the sequence of three profiles focused on subsequent (partially overlapped) frames extracted from the collected signal. An ISAR CPI of 6 s is used, sliding along the 10 s of the globally available acquisition time. The different frames refer to image time spanning from 4 to 6 s with spacing 1 s. We observe that two almost identical cross-range profiles are obtained: these profiles persist in all the three considered frames occupying progressively increasing cross-range positions, with a relative displacement

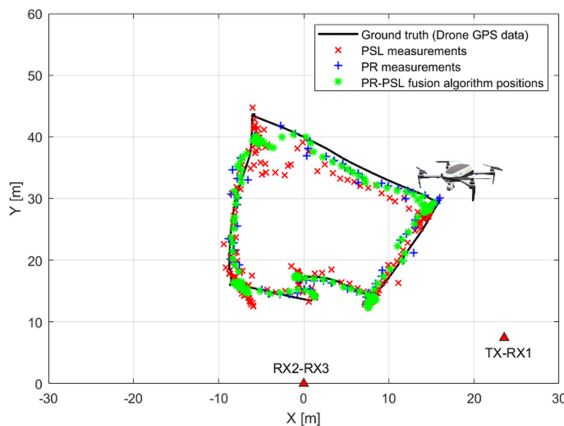


Figure 14.

Drone detection results with PR, PSL, and PR-PSL fusion algorithm. Reprinted, with permission, from [60].

between the first and the last frame of about 8.7 m, which is well in line with the target cross-range velocity.

Moreover, we observe that each of the two identical patterns in the reported cross-range profiles is characterized by three main peaks, which correspond to the main scattering centers of the two identical cars used for the considered experiment. This clearly shows that the proposed technique allowed us both to separate the two vehicles in the cross-range direction and to identify their main scattering points, making it possible to measure their displacement. This represents the first step for a target classification procedure since it gives an indication of the target electromagnetic length and signature. Enhanced imaging/classification capability could be expected by exploiting multistatic configurations, thanks to the diversity provided by sparse sensors distribution.

JOINT PASSIVE RADAR AND PASSIVE LOCATION

As previously mentioned, the use of signals of opportunity in the WiFi band potentially enables PR and device-based passive source location (PSL) techniques to be jointly exploited against the data collected by a unique receiver. In fact, many targets of interest for the applications at hand typically carry devices that emit signals in the WiFi band. This is, for example, the case of smartphones carried by people or WiFi transceivers onboard of commercial drones. In such cases, information about the presence of a target and its position can be obtained through its direct emissions.

In [60], it has been shown that the joint exploitation of the signals backscattered by the target's body according to a PR approach and the signals emitted by WiFi devices according to a PSL approach allows us to improve the capability to detect and localize the targets, thanks to both the augmentation of the available observations and the complementarity of the two approaches. In contrast to the PR, the PSL requires the target to be equipped with a WiFi device; thus, it might be unsuitable for security applications.

Also, it could be potentially inaccurate for moving targets since the use of WiFi networks facilities is expected to reduce when the target moves. On the other hand, PSL is an interesting solution for stationary objects localization, which is not guaranteed by the PR since their echoes are canceled together with the direct signal and its multipath. Moreover, PSL allows the unambiguous association of the transmission to a specific target, based on the device MAC address, so that even very closely spaced targets can be discriminated. By exploiting one-way signals, the PSL typically relies on higher SNR conditions, which provide the possibility to reduce the complexity of the processing scheme, since it does not require all the steps needed by the PR for the extraction of the target echoes.

The joint exploitation of PR and PSL approaches was tested on experimental data collected in dedicated acquisition campaigns [60]. A DJI Mavic Pro drone (size: $30 \times 25 \times 8$ cm, weight: 730 g) was used as a cooperative target, which guarantees WiFi 802.11a/b/g/n/ac connectivity. The streaming video from drone to controller was exploited as device transmissions to perform target detection and localization according to a PSL approach. Drone localization was performed by means of a proper combination of bistatic range and angle measurements for the PR sensor while the PSL exploits the time difference of arrival (TDoA) and the angle of arrival (AoA) measurement to accomplish the same task. In the example reported here, the drone moves along a quadrangular trajectory and it stops for a few seconds at each vertex (the black line in Figure 14 reports the ground truth obtained from drone GPS recording). The localization results separately obtained with the PSL and the PR approaches are reported in Figure 14 using red and blue markers, respectively. Both the reported curves approximate with reasonable accuracy the drone's trajectory in the X - Y plane. However, one can verify that the PR sensor does not provide a full coverage of the drone flight, as it fails at detecting its presence in the time lapses when the target is in hovering. In contrast, the localization results obtained with the PSL approach might be inaccurate when the WiFi packets emission rate from the drone to the controller reduces whereas the PR provides a good accuracy due to the higher transmission rate of the AP.

In Figure 14, with green markers, we also report the results of a fusion algorithm operated at the tracking stage. As is apparent, the green track yields the best performance in terms of coverage and accuracy since it benefits from the advantages carried by the two techniques jointly exploited.

FUTURE OUTLOOK AND PERSPECTIVES

In many of the applications reported in this article, PR has reached a point of maturity with several prototypes and systems available in the market.

However, the research activity on PR technology has not stopped indeed and many aspects are currently being

investigated with the aim to provide these sensors with advanced capabilities and to extend their range of uses. It is therefore reasonable to ask what the current trends and future perspectives for this technology are.

These certainly include the possibility of installing a PR onboard of a moving platform since this yields a number of strategic advantages if the specific technological and methodological challenges of this application are successfully tackled [24], [63], [64]. Among these challenges, the most important ones are the size and the weight of the system, especially when small platforms are considered, but also the effects that the platform motion has on the PR operation. These include the need to recover the reference signal from the moving platform, but also the clutter Doppler spectrum spreading due to the platform motion, which causes slowly moving target echoes to be masked by the returns from the stationary scene.

Among the advantages, the possibility to install a PR on a moving platform could offer covert and low-cost monitoring of wide areas and implement an easy deployable surveillance system, for example, if we think about installing a PR on board of an UAV. It enables advanced modes, such as SAR imaging and ground moving target indication. In addition, it is worth mentioning its potentialities as autonomous driving assistance system (ADAS). In fact, having different radar units to share the same transmitted signal, the PR concept inherently eliminates the problem of mutual interference, which might be a major safety concern for active ADAS, especially in congested areas. Considering this advantage, together with the additional benefits of reduced costs and energy consumption, there might be a great interest in using a PR in automotive applications, at least as a backup system.

The abovementioned opportunities are further enhanced by the availability of a multitude of transmitters of opportunity, which are proliferating for various communications applications. Among them, the ground-based transmitters of future generations of technology standard for cellular networks (e.g., 5G and 6G) and the constellations of satellite transmitters providing global internet access services (e.g., Starlink) play a key role since these systems make available an extremely high number of IOs thus possibly covering any given point on the Earth surface [65], [66], [67].

With particular reference to the use of satellite illuminators, we have shown in this article that this allows a number of advanced applications, such as maritime/river surveillance in remote areas, the protection of infrastructures, with low vulnerability, the SAR imaging, and so on. Obviously, there are several challenges to be faced when exploiting a satellite transmitter. First, the very low power density that is available on the Earth surface. This especially applies to the case of the exploitation of geostationary satellites, which are thousands of kilometers away from the Earth surface. Also, the need to track the satellites and fruitfully exploit their Doppler-shifted signal might represent an issue when instead satellites in the medium and low Earth orbits are exploited.

This not only requires advanced processing strategies but also flexible hardware solutions and reconfigurable antennas, e.g., based on metamaterials.

The coverage of the multistatic system could be widened by developing and deploying a dense network of low-cost receivers properly displaced across the area to be surveyed. Key points for such approach would be the capability to synchronize the different nodes of the network especially in GPS-denied environment (e.g., indoor, defence applications) as well as the need to keep the cost of the resulting system low. To this aim, when extreme bistatic configurations are used or for very short-range surveillance applications, it is possible to resort to forward scatter radar approaches [68], [69], [70]. In fact, these require a noncoherent signal processing strategy, thus relaxing the requirements in terms of phase synchronization and signal processing complexity.

Being based on the parasitic exploitation of transmitters for communications services, PR technology can be regarded as a pioneering form of co-habitation between radar and communication systems where the radar functionality is totally subject to the design of the communications system and the radar component operates in the worst conditions [71], [72]. Indeed, improved radar performance can be achieved if at least a partial cooperation is implemented between the communications and the radar systems sharing the same spectral and hardware resources. This aspect has come to be a primary investigation field in recent years and has attracted tremendous attention from both academia and industry. It is motivated by the proliferation of wireless communications and radio-navigation services that has determined an increasing load on finite spectral resources but also by the need for advanced services and applications that in turn require the system to be equipped with “sensing” capabilities. Still, there are numerous technical challenges to be solved in order to enable the spectrum sharing across disparate radio frequency users that are typically characterized by different requirements and competing performance. Moreover, there is scope to reveal specific strategies for spectrum sharing, ranging from co-existence to co-design, and this is expected to stimulate new contributions in terms of technical innovation, standardization, and novel applications.

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REFERENCES

- [1] S. S. Swords, *Technical History of the Beginning of Radar*. London, U.K.: IET, 1986.
- [2] H. Kuschel, "Approaching 80 years of passive radar," in *Proc. Int. Conf. Radar*, 2013, pp. 213–217.
- [3] H. Griffiths, "Early history of bistatic radar," in *Proc. Eur. Radar Conf.*, 2016, pp. 253–257, doi: [10.1109/RADAR.2013.6651987](https://doi.org/10.1109/RADAR.2013.6651987).
- [4] H. Kuschel, D. Cristallini, and K. E. Olsen, "Tutorial: Passive radar tutorial," *IEEE Aerosp. Electron. Syst. Mag.*, vol. 38, no. 2, pp. 2–19, Feb. 2019, doi: [10.1109/MAES.2018.160146](https://doi.org/10.1109/MAES.2018.160146).
- [5] A. Farina, *Green Radar State of Art: Theory, Practice and Way Ahead* (Sep. 8, 2015). [Online Video]. Available: <https://ieeetv.ieee.org/green-radar-state-of-art-theory-practice-and-way-ahead>
- [6] H. Griffiths and N. Willis, "Klein Heidelberg—The first modern bistatic radar system," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 46, no. 4, pp. 1571–1588, Oct. 2010, doi: [10.1109/TAES.2010.5595580](https://doi.org/10.1109/TAES.2010.5595580).
- [7] J. Gough, *Watching The Skies—The History of Ground Radar in the Air Defence of the United Kingdom*. London, U.K.: HMSO, 1993.
- [8] H. D. Griffiths and N. R. W. Long, "Television-based bistatic radar," *IEE Proc. F—Commun., Radar Signal Process.*, vol. 133, pp. 649–657, Dec. 1986, doi: [10.1049/ip-f.1.1986.0104](https://doi.org/10.1049/ip-f.1.1986.0104).
- [9] P. E. Howland, "Target tracking using television-based bistatic radar," *IEE Proc.—Radar, Sonar Navigation*, vol. 146, pp. 166–174, Jun. 1999, doi: [10.1049/ip-rsn:19990322](https://doi.org/10.1049/ip-rsn:19990322).
- [10] H. Griffiths et al., "Radar spectrum engineering and management: Technical and regulatory issues," *Proc. IEEE*, vol. 103, no. 1, pp. 85–102, Jan. 2015, doi: [10.1109/JPROC.2014.2365517](https://doi.org/10.1109/JPROC.2014.2365517).
- [11] H. Griffiths and C. Baker, "Passive coherent location radar systems. Part 1: Performance prediction," *Proc. Inst. Elect. Eng. Radar Sonar Navigation*, vol. 152, pp. 153–159, 2005, doi: [10.1049/ip-rsn:20045082](https://doi.org/10.1049/ip-rsn:20045082).
- [12] C. J. Baker, H. D. Griffiths, and I. Papoutsis, "Passive coherent location radar systems. Part 2: Waveform properties," *IEE Proc. Radar Sonar Navigation*, vol. 152, pp. 160–168, 2005, doi: [10.1049/ip-rsn:20045083](https://doi.org/10.1049/ip-rsn:20045083).
- [13] P. E. Howland, D. Maksimiuk, and G. Reitsma, "FM radio based bistatic radar," *IEE Proc. Radar Sonar Navigation*, vol. 152, pp. 107–115, 2005, doi: [10.1049/ip-rsn:20045077](https://doi.org/10.1049/ip-rsn:20045077).
- [14] C. Coleman and H. Yardley, "Passive bistatic radar based on target illuminations by digital audio broadcasting," *IET Radar Sonar Navigation*, vol. 2, no. 5, Oct. 2008, doi: [10.1049/iet-rsn:20080019](https://doi.org/10.1049/iet-rsn:20080019).
- [15] "Special issue on passive radar I & II," *IEEE Aerosp. Electron. Syst. Mag.*, vol. 27, no. 10, 2012.
- [16] H. Griffiths and C. Baker, "Passive bistatic radar," in *Principles of Modern Radar. Vol. III, Radar Applications*, W. L. Melvin and J. A. Scheer, Eds. Raleigh, NC, USA: SciTech Publishing, 2014, pp. 499–541.
- [17] H. D. Griffiths and C. J. Baker, *An Introduction to Passive Radar*. Norwood, MA, USA: Artech House, Feb. 2017.
- [18] R. Klemm Eds., "Novel radar techniques and applications," in *Part III: Passive and Multistatic Radar*. Stevenage, U.K.: IET, 2017.
- [19] M. Malanowski, *Signal Processing for Passive Bistatic Radar*. Norwood, MA, USA: Artech House, 2019.
- [20] B. D. Nordwall, "'Silent sentry' a new type of radar," *Aviation Week Space Technol.*, vol. 149, pp. 70–71, Nov. 1998.
- [21] F. Colone, D. W. O'Hagan, P. Lombardo, and C. J. Baker, "A multistage processing algorithm for disturbance removal and target detection in passive bistatic radar," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 45, no. 2, pp. 698–722, Apr. 2009, doi: [10.1109/TAES.2009.5089551](https://doi.org/10.1109/TAES.2009.5089551).
- [22] P. Lombardo and F. Colone, "Advanced processing methods for passive bistatic radar," in *Principles of Modern Radar: Advanced Radar Techniques*, W. L. Melvin and J. A. Scheer, Eds. Raleigh, NC, USA: SciTech Publishing, 2012, pp. 739–821.
- [23] F. Colone, C. Palmari, T. Martelli, and E. Tilli, "Sliding extensive cancellation algorithm for disturbance removal in passive radar," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 52, no. 3, pp. 1309–1326, Jun. 2016, doi: [10.1109/TAES.2016.150477](https://doi.org/10.1109/TAES.2016.150477).
- [24] P. Wojaczek, F. Colone, D. Cristallini, and P. Lombardo, "Reciprocal-filter-based STAP for passive radar on moving platforms," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 55, no. 2, pp. 967–988, Apr. 2019, doi: [10.1109/TAES.2018.2867688](https://doi.org/10.1109/TAES.2018.2867688).
- [25] D. Reid, "An algorithm for tracking multiple targets," *IEEE Trans. Autom. Control*, vol. 24, no. 6, pp. 843–854, Dec. 1979, doi: [10.1109/TAC.1979.1102177](https://doi.org/10.1109/TAC.1979.1102177).
- [26] S. Blackman and R. Popoli, *Design and Analysis of Modern Tracking Systems*. Norwood, MA, USA: Artech House, 1999.
- [27] F. Colone and P. Lombardo, "Polarimetric passive coherent location," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 51, no. 2, pp. 1079–1097, Apr. 2015, doi: [10.1109/TAES.2014.130775](https://doi.org/10.1109/TAES.2014.130775).
- [28] F. Filippini, F. Colone, D. Cristallini, and G. Bournaka, "Experimental results of polarimetric detection schemes for DVB-T-based passive radar," *IET Radar, Sonar Navigation*, vol. 11, no. 6, pp. 883–891, 2017, doi: [10.1049/iet-rsn.2016.0486](https://doi.org/10.1049/iet-rsn.2016.0486).
- [29] F. Filippini and F. Colone, "Polarimetric passive radar: A practical approach to parametric adaptive detection," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 56, no. 6, pp. 4930–4946, Dec. 2020, doi: [10.1109/TAES.2020.3008548](https://doi.org/10.1109/TAES.2020.3008548).

- [30] F. Colone, C. Bongioanni, and P. Lombardo, "Multi-frequency integration in FM radio based passive bistatic radar. Part I: Target detection," *IEEE Aerosp. Electron. Syst. Mag.*, vol. 28, no. 4, pp. 28–39, Apr. 2013, doi: [10.1109/MAES.2013.6506827](https://doi.org/10.1109/MAES.2013.6506827).
- [31] F. Colone, C. Bongioanni, and P. Lombardo, "Multi-frequency integration in FM radio based passive bistatic radar. Part II: Direction of arrival estimation," *IEEE Aerosp. Electron. Syst. Mag.*, vol. 28, no. 4, pp. 40–47, Apr. 2013, doi: [10.1109/MAES.2013.6506828](https://doi.org/10.1109/MAES.2013.6506828).
- [32] T. Martelli, F. Colone, E. Tilli, and A. Di Lallo, "Multi-frequency target detection techniques for DVB-T based passive radar sensors," *Sensors*, vol. 16, pp. 1594, 2016, doi: [10.3390/s16101594](https://doi.org/10.3390/s16101594).
- [33] F. Colone and P. Lombardo, "Non-coherent adaptive detection in passive radar exploiting polarimetric and frequency diversity," *IET Radar, Sonar Navigation*, vol. 10, no. 1, pp. 15–23, 2016, doi: [10.1049/iet-rsn.2015.0104](https://doi.org/10.1049/iet-rsn.2015.0104).
- [34] F. Filippini and F. Colone, "Multi-carrier adaptive detection in polarimetric passive radars," in *Proc. IEEE Radar Conf.*, 2020, pp. 1–6, doi: [10.1109/RadarConf2043947.2020.9266347](https://doi.org/10.1109/RadarConf2043947.2020.9266347).
- [35] F. Filippini and F. Colone, "Multi-carrier and multi-polarimetric model based adaptive target detector for passive radar systems," *IET Radar, Sonar Navigation*, vol. 15, no. 8, pp. 853–866, Apr. 2021, doi: [10.1049/rsn2.12061](https://doi.org/10.1049/rsn2.12061).
- [36] A. Di Lallo et al., "AULOS: Finmeccanica family of passive sensors," *IEEE Aerosp. Electron. Syst. Mag.*, vol. 31, no. 11, pp. 24–29, Nov. 2016, doi: [10.1109/MAES.2017.160037](https://doi.org/10.1109/MAES.2017.160037).
- [37] T. Martelli, F. Colone, and R. Cardinali, "DVB-T based passive radar for simultaneous counter drone operations and civil air traffic surveillance," *IET Radar, Sonar Navigation*, vol. 14, no. 4, pp. 505–515, 2020, doi: [10.1049/iet-rsn.2019.0309](https://doi.org/10.1049/iet-rsn.2019.0309).
- [38] P. Lombardo et al., *Multiband Passive Radar For Drones Detection and Localization (Radar, Sonar and Navigation, 2021), Radar Countermeasures For Unmanned Aerial Vehicles*. London, U.K.: IET, 2021, pp. 213–250.
- [39] M. Antoniou and M. Cherniakov, "GNSS-based passive radar," in *Novel Radar Techniques and Applications, Part III: Passive and Multistatic Radar*, R. Klemm, Ed. London, U.K.: IET, 2017.
- [40] F. Santi, F. Pieralice, and D. Pastina, "Joint detection and localization of vessels at sea with a GNSS-based multistatic radar," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 8, pp. 5894–5913, Aug. 2019, doi: [10.1109/TGRS.2019.2902938](https://doi.org/10.1109/TGRS.2019.2902938).
- [41] F. Santi, D. Pastina, and M. Bucciarelli, "Experimental demonstration of ship target detection in GNSS-based passive radar combining target motion compensation and track-before-detect strategies," *Sensors*, vol. 20, no. 3, 2020, Art. no. 599, doi: [10.3390/s20030599](https://doi.org/10.3390/s20030599).
- [42] H. Ma et al., "Maritime moving target indication using passive GNSS-based bistatic radar," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 54, no. 1, pp. 115–130, Feb. 2018, doi: [10.1109/TAES.2017.2739900](https://doi.org/10.1109/TAES.2017.2739900).
- [43] D. Pastina et al., "Maritime moving target long time integration for GNSS-based passive bistatic radar," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 54, no. 6, pp. 3060–3083, Dec. 2018, doi: [10.1109/TAES.2018.2840298](https://doi.org/10.1109/TAES.2018.2840298).
- [44] D. Pastina, F. Santi, F. Pieralice, M. Antoniou, and M. Cherniakov, "Passive radar imaging of ship targets with GNSS signals of opportunity," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 3, pp. 2627–2642, Mar. 2021, doi: [10.1109/TGRS.2020.3005306](https://doi.org/10.1109/TGRS.2020.3005306).
- [45] I. Pisciotano, D. Cristallini, and D. Pastina, "Maritime target imaging via simultaneous DVB-T and DVB-S passive ISAR," *IET Radar, Sonar Navigation*, vol. 13, no. 9, pp. 1479–1487, Sep. 2019, doi: [10.1049/iet-rsn.2018.5622](https://doi.org/10.1049/iet-rsn.2018.5622).
- [46] F. Santi, D. Pastina, M. Antoniou, and M. Cherniakov, "GNSS-based multistatic passive radar imaging of ship targets," in *Proc. IEEE Int. Radar Conf.*, 2020, pp. 601–606, doi: [10.1109/RADAR42522.2020.9114638](https://doi.org/10.1109/RADAR42522.2020.9114638).
- [47] F. Santi, I. Pisciotano, D. Pastina, and D. Cristallini, "First experimental results on multi-angle DVB-S based passive ISAR exploiting multipolar data," in *Proc. IEEE Radar Conf.*, 2021, pp. 1–6, doi: [10.1109/RadarConf2147009.2021.9455209](https://doi.org/10.1109/RadarConf2147009.2021.9455209).
- [48] I. Pisciotano, F. Santi, D. Pastina, and D. Cristallini, "DVB-S based passive polarimetric ISAR—Methods and experimental validation," *IEEE Sensors J.*, vol. 21, no. 5, pp. 6056–6070, Mar. 2021, doi: [10.1109/JSEN.2020.3037091](https://doi.org/10.1109/JSEN.2020.3037091).
- [49] O. Cabrera, C. Bongioanni, F. Filippini, O. Sarabakha, F. Colone, and P. Lombardo, "Detecting drones and human beings with DVB-S based COTS passive radar for short-range surveillance," in *Proc. IEEE Int. Radar Conf.*, 2020, pp. 37–42, doi: [10.1109/RADAR42522.2020.9114795](https://doi.org/10.1109/RADAR42522.2020.9114795).
- [50] F. Filippini, O. Cabrera, C. Bongioanni, F. Colone, and P. Lombardo, "DVB-S based passive radar for short range security application," in *Proc. IEEE Radar Conf.*, 2021, pp. 1–6, doi: [10.1109/RadarConf2147009.2021.9455242](https://doi.org/10.1109/RadarConf2147009.2021.9455242).
- [51] F. Colone, K. Woodbridge, H. Guo, D. Mason, and C. J. Baker, "Ambiguity function analysis of wireless LAN transmissions for passive radar," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 47, no. 1, pp. 240–264, Jan. 2011, doi: [10.1109/TAES.2011.5705673](https://doi.org/10.1109/TAES.2011.5705673).
- [52] F. Colone, P. Falcone, C. Bongioanni, and P. Lombardo, "WiFi-based passive bistatic radar: Data processing schemes and experimental results," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 48, no. 2, pp. 1061–1079, Apr. 2012, doi: [10.1109/TAES.2012.6178049](https://doi.org/10.1109/TAES.2012.6178049).
- [53] K. Chetty, G. E. Smith, and K. Woodbridge, "Through-the-wall sensing of personnel using passive bistatic WiFi radar at standoff distances," *IEEE Trans. Geosci. Remote Sens.*, vol. 50, no. 4, pp. 1218–1226, Apr. 2012, doi: [10.1109/TGRS.2011.2164411](https://doi.org/10.1109/TGRS.2011.2164411).
- [54] P. Falcone, F. Colone, and P. Lombardo, "Potentialities and challenges of WiFi-based passive radar," *IEEE Aerosp. Electron. Syst. Mag.*, vol. 27, no. 11, pp. 15–26, Nov. 2012, doi: [10.1109/MAES.2012.6380822](https://doi.org/10.1109/MAES.2012.6380822).

- [55] F. Colone, D. Pastina, P. Falcone, and P. Lombardo, "WiFi-based passive ISAR for high-resolution cross-range profiling of moving targets," *IEEE Trans. Geosci. Remote Sens.*, vol. 52, no. 6, pp. 3486–3501, Jun. 2014, doi: [10.1109/TGRS.2013.2273099](https://doi.org/10.1109/TGRS.2013.2273099).
- [56] D. Pastina, F. Colone, T. Martelli, and P. Falcone, "Parasitic exploitation of Wi-Fi signals for indoor radar surveillance," *IEEE Trans. Veh. Technol.*, vol. 64, no. 4, pp. 1401–1415, Apr. 2015, doi: [10.1109/TVT.2015.2392936](https://doi.org/10.1109/TVT.2015.2392936).
- [57] F. Colone, "Short-range passive radar potentialities," in *Novel Radar Techniques and Applications Volume 1: Real Aperture Array Radar, Imaging Radar, and Passive and Multistatic Radar*, R. Klemm, Ed. London, U.K.: IET, 2017.
- [58] Y. L. Q.Chen, B. Tan, K. Woodbridge, and K. Chetty, "Respiration and activity detection based on passive radio sensing in home environments," *IEEE Access*, vol. 8, pp. 12426–12437, 2020, doi: [10.1109/ACCESS.2020.2966126](https://doi.org/10.1109/ACCESS.2020.2966126).
- [59] W. Li, R. J. Piechocki, K. Woodbridge, C. Tang, and K. Chetty, "Passive WiFi radar for human sensing using a stand-alone access point," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 3, pp. 1986–1998, Mar. 2021, doi: [10.1109/TGRS.2020.3006387](https://doi.org/10.1109/TGRS.2020.3006387).
- [60] I. Milani, C. Bongioanni, F. Colone, and P. Lombardo, "Fusing measurements from Wi-Fi emission-based and passive radar sensors for short-range surveillance," *Remote Sens.*, vol. 13, 2021, Art. no. 3556, doi: [10.3390/rs13183556](https://doi.org/10.3390/rs13183556).
- [61] H. Sun, L. G. Chia, and S. G. Razul, "Through-wall human sensing with WiFi passive radar," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 57, no. 4, pp. 2135–2148, Aug. 2021, doi: [10.1109/TAES.2021.3069767](https://doi.org/10.1109/TAES.2021.3069767).
- [62] F. Colone, F. Filippini, M. Di Seglio, and K. Chetty, "On the use of reciprocal filter against WiFi packets for passive radar," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 58, no. 4, pp. 2746–2761, Aug. 2022, doi: [10.1109/TAES.2021.3138711](https://doi.org/10.1109/TAES.2021.3138711).
- [63] G. P. Blasone, F. Colone, P. Lombardo, P. Wojaczek, and D. Cristallini, "Passive radar DPCA schemes with adaptive channel calibration," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 56, no. 5, pp. 4014–4034, Oct. 2020, doi: [10.1109/TAES.2020.2987478](https://doi.org/10.1109/TAES.2020.2987478).
- [64] G. Mazurek, K. Kulpa, and A. Droszcz, "Multichannel SDR platforms for mobile PCL applications," *IEEE Aerosp. Electron. Syst. Mag.*, vol. 37, no. 4, pp. 30–41, Apr. 2022, doi: [10.1109/MAES.2021.3052307](https://doi.org/10.1109/MAES.2021.3052307).
- [65] P. Samczyński et al., "5G network-based passive radar," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5108209, doi: [10.1109/TGRS.2021.3137904](https://doi.org/10.1109/TGRS.2021.3137904).
- [66] A. Sayin, M. Cherniakov, and M. Antoniou, "Passive radar using starlink transmissions: A theoretical study," in *Proc. 20th Int. Radar Symp.*, 2019, pp. 1–7, doi: [10.23919/IRS.2019.8768105](https://doi.org/10.23919/IRS.2019.8768105).
- [67] P. Gomez-Del-Hoyo, K. Gronowski, and P. Samczynski, "The STARLINK-based passive radar: Preliminary study and first illuminator signal measurements," in *Proc. 23rd Int. Radar Symp.*, 2022, pp. 350–355, doi: [10.23919/IRSS4158.2022.9905046](https://doi.org/10.23919/IRSS4158.2022.9905046).
- [68] M. Cherniakov, "Basic principles of forward-scattering radars," in *Bistatic Radar: Principles and Practice: Part III*. New York, NY, USA: Wiley, 2007.
- [69] A. Arcangeli, C. Bongioanni, N. Ustalli, D. Pastina, and P. Lombardo, "Passive forward scatter radar based on satellite TV broadcast for air target detection: Preliminary experimental results," in *Proc. IEEE Radar Conf.*, 2017, pp. 1592–1596, doi: [10.1109/RADAR.2017.7944462](https://doi.org/10.1109/RADAR.2017.7944462).
- [70] F. Colone, "DVB-T-based passive forward scatter radar: Inherent limitations and enabling solutions," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 57, no. 2, pp. 1084–1104, Apr. 2021, doi: [10.1109/TAES.2020.3035435](https://doi.org/10.1109/TAES.2020.3035435).
- [71] Y. Cui, F. Liu, X. Jing, and J. Mu, "Integrating sensing and communications for ubiquitous IoT: Applications, trends, and challenges," *IEEE Netw.*, vol. 35, no. 5, pp. 158–167, Sep./Oct. 2021, doi: [10.1109/MNET.010.2100152](https://doi.org/10.1109/MNET.010.2100152).
- [72] A. Liu et al., "A survey on fundamental limits of integrated sensing and communication," *IEEE Commun. Surv. Tut.*, vol. 24, no. 2, pp. 994–1034, Apr.–Jun. 2022, doi: [10.1109/COMST.2022.3149272](https://doi.org/10.1109/COMST.2022.3149272).